

### **Exp No: 15**

#### **EMBEDDED C PROGRAM TO CONVERT THE HEXADECIMAL DATA 0XCFH TO DECIMAL AND DISPLAY THE DIGITS ON PORTS P0, P1 AND P2**

##### **AIM:**

To write an Embedded C program to convert Hexadecimal numbers into Decimal and display the digits on ports using Keil software.

##### **PROGRAM:**

```
#include <reg51.h>

void main(void)
{
    unsigned char hexa = 0x08;
    unsigned char hundreds, tens, units;

    ACC = hexa;
    B = 10;
    ACC = ACC / B;
    units = B;

    B = 10;

    ACC = ACC / B;
    tens = B;
    hundreds = ACC;

    P0 = units;
    P1 = tens;
    P2 = hundreds;

    while(1);
}
```

```
6: unsigned char hundreds, tens, units;
```

Parallel Port 0

Port 0

P0: 0x07

7 Bits 0

Pins: 0x07

7 Bits 0

Parallel Port 1

Port 1

P1: 0x00

7 Bits 0

Pins: 0x00

7 Bits 0

Parallel Port 2

Port 2

P2: 0x02

7 Bits 0

Pins: 0x02

7 Bits 0

hex  
hun

al

temp;

et the Units digit

= 20 (Quotient in temp), Remain

10;

% 10;

15

// Step 2: Get the Tens and Hundreds digits